Establish a central limit theorem for stochastic mirror descent risk budgeting portfolios

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Website #2 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

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Problem

Let

$$z = (\xi, y) \in \mathbb{R} \times (0,\infty)^d, \qquad H(z,X) = L(\xi, -\langle y, X\rangle) - \sum_{i=1}^{d} b_i \log y_i,$$

and

$$h(z) = \mathbb{E}[H(z,X)].$$

The standing assumptions imply that $z^\star = (\xi^\star, y^\star)$ is the unique minimizer of h, with $y^\star \in (0,\infty)^d$, and the stochastic mirror descent (SMD) recursion is

$$\xi_{n+1} = \xi_n - \gamma_{n+1}\,\partial_\xi H(z_n, X_{n+1}),$$

$$y_{n+1} = P^m_{y_n}(\gamma_{n+1}\kappa(y_n)\nabla_y H(z_n, X_{n+1}))\,, \qquad \kappa(y) := \Big(\min_i y_i\Big) \wedge 1.$$

The question asks for an asymptotic distributional characterization of the SMD risk-budgeting estimator: find a deterministic normalization $a_n \to \infty$ and a covariance matrix Σ such that either the raw iterates or the averaged iterates satisfy a CLT of the form

$$a_n(z_n - z^\star) \Rightarrow \mathcal{N}(0,\Sigma) \quad \text{or} \quad a_n(\bar{z}_n - z^\star) \Rightarrow \mathcal{N}(0,\Sigma).$$

I will prove a rigorous CLT for the *raw iterates* under additional interiority, local smoothness, and moment assumptions. I also derive the induced CLT for the normalized portfolio weights

$$u_n := \frac{y_n}{\|y_n\|_1}, \qquad u^\star := \frac{y^\star}{\|y^\star\|_1}.$$

Alternative readings considered

- The prompt explicitly allows a CLT for either the raw iterates z_n or the averaged iterates $\bar{z}_n$. I choose the raw iterates.
- The hardest reading would allow the radius constraint to remain asymptotically active when $m = \|y^\star\|_1$. I do *not* solve that case; I retreat to the interior case $m > \|y^\star\|_1$, where the KL projection is eventually inactive.
- The standing assumptions only give C^1-regularity for the deterministic objective. For a genuine CLT one needs more. I therefore add explicit C^2-type local regularity and moment assumptions.

Related literature found

- **Arnaiz Iglesias–Cetingoz–Frikha (2024),** `BACKGROUND`**.** This is the exact preprint behind the prompt: it proves almost sure convergence and a non-asymptotic bound for the averaged mirror-descent scheme, but does not provide a CLT.

- **Polyak–Juditsky (1992),** `PARTIAL/TOOL`**.** Classical trajectory-averaging CLT for stochastic approximation. It strongly suggests that averaging should yield a Gaussian limit, but their averaging is ordinary Cesàro averaging, not the $\sum \gamma_k$-weighted average in the prompt.

- **Fabian (1968), Hernandez–Spall (2019),** `TOOL`**.** These are canonical raw-iterate SA CLT references. They indicate the correct Lyapunov covariance equation and the spectral threshold $1/2$, but I do not import them as black boxes.

- **Buche–Kushner (2002),** `PARTIAL`**.** This paper explains why boundary-active constrained stochastic approximation typically leads to reflected linear diffusions rather than the unconstrained Gaussian limit. This is precisely why I impose $m > \|y^\star\|_1$.

- **Cutler–Diaz–Drusvyatskiy (2024),** `PARTIAL/TOOL`**.** Modern open-access averaged-SA CLT in a nearby Euclidean projected setting. It is informative, but it does not directly cover KL mirror steps with the weighted average from the prompt.

- **Zhu (1996),** `TOOL`**.** The abstract matches almost exactly the linearized recursion obtained below, but I did not rely on it because I did not audit the full theorem statement and hypotheses from an accessible full text.

Proof sketch

The key point is that the mirror mechanism is only genuinely non-Euclidean away from the optimum. Near $z^\star$, and under the interior condition $m > \|y^\star\|_1$, the KL projection becomes eventually inactive and the y-update is exactly the multiplicative update

$$y_{n+1} = y_n \odot \exp\bigl(-\gamma_{n+1}\kappa(y_n)\nabla_y H(z_n, X_{n+1})\bigr).$$

A first-order Taylor expansion of the exponential shows that, locally, the recursion is an ordinary stochastic approximation scheme with deterministic preconditioner

$$G_\star = \mathrm{diag}\bigl(1, \kappa_\star y_1^\star, \ldots, \kappa_\star y_d^\star\bigr), \qquad \kappa_\star := \kappa(y^\star).$$

Thus the mirror map and the taming function enter the CLT only through this matrix $G_\star$.

The deterministic drift is

$$F(z) = G(z)\nabla h(z).$$

Because $\nabla h(z^\star) = 0$, the possible nonsmoothness of κ at $y^\star$ is harmless at first order: the linearization is simply

$$DF(z^\star) = G_\star \nabla^2 h(z^\star).$$

Hence the linearized dynamics is

$$e_{n+1} \approx \Bigl(I - \frac{\gamma}{n+1}A\Bigr)e_n - \frac{\gamma}{n+1}\zeta_{n+1}, \qquad A := G_\star \nabla^2 h(z^\star),$$

with centered i.i.d. noise

$$\zeta_{n+1} = G_\star \nabla H(z^\star, X_{n+1}).$$

I then prove three things.

First, the nonlinear mirror recursion differs from that linear SA recursion by a deterministic remainder of order n^{-2} and a martingale remainder whose variance is negligible at the $\sqrt{n}$-scale.

Second, for the linear recursion I write the explicit product formula and obtain the asymptotics

$$\prod_{j=k+1}^{n} \Big(I - \frac{\gamma A}{j}\Big) = \Big(\frac{k}{n}\Big)^{\gamma A} \big(I + o(1)\big),$$

which yields the exact covariance candidate

$$\Sigma = \gamma^2 \int_0^\infty e^{-s(\gamma A - \frac{1}{2} I)} Q \, e^{-s(\gamma A^\top - \frac{1}{2} I)} \, ds, \qquad Q := \mathrm{Cov}(\zeta_1).$$

Third, I prove a triangular-array CLT for the explicit linear representation by a characteristic-function computation, avoiding any external black-box SA CLT. The condition

$$\min \Re \lambda(\gamma A) > \frac{1}{2}$$

is exactly what makes the above integral finite and yields the $\sqrt{n}$-normalization.

Main result

Theorem 1 (Interior raw-iterate CLT for the SMD risk-budgeting estimator)**.** *Let $p = d + 1$, let*

$$z_n = (\xi_n, y_n), \qquad e_n := z_n - z^\star,$$

and define

$$G(z) := \begin{pmatrix} 1 & 0 \\ 0 & \kappa(y)\,\mathrm{Diag}(y) \end{pmatrix}, \qquad \Psi(z, x) := G(z) \nabla_z H(z, x).$$

Assume, in addition to the standing hypotheses of the problem, that:

(A1) the step sizes are $\gamma_n = \gamma / n$ with $\gamma > 0$;

(A2) $m > \|y^\star\|_1$;

(A3) there exists a neighborhood U of $z^\star$ such that h is twice continuously differentiable on U, with positive-definite Hessian

$$S := \nabla^2 h(z^\star) \in \mathbb{R}^{p \times p};$$

(A4) there exist random variables $M_0, M_1 \in L^3$ such that, almost surely, for all $z, z' \in U$,

$$\|\nabla_z H(z, X_1)\| \le M_0, \qquad \|\nabla_z H(z, X_1) - \nabla_z H(z', X_1)\| \le M_1 \|z - z'\|;$$

(A5) the recursion converges almost surely to $z^\star$, and moreover

$$\sup_{n \ge 1} n \, \mathbb{E} \|e_n\|^2 < \infty;$$

(A6) with

$$\kappa_\star := \kappa(y^\star), \qquad G_\star := G(z^\star) = \mathrm{diag}(1, \kappa_\star y_1^\star, \dots, \kappa_\star y_d^\star),$$

the matrix

$$A := G_\star S$$

satisfies

$$\beta := \min \Re\lambda(\gamma A) > \frac{1}{2}.$$

Then

$$\sqrt{n}\,(z_n - z^\star) \Rightarrow \mathcal{N}(0, \Sigma),$$

where $Q := \mathrm{Cov}(\zeta_1)$ *with*

$$\zeta_1 := \Psi(z^\star, X_1) = G_\star \nabla_z H(z^\star, X_1),$$

and Σ *is the unique solution of the Lyapunov equation*

$$\Big(\gamma A - \frac{1}{2} I_p\Big)\Sigma + \Sigma\Big(\gamma A^\top - \frac{1}{2} I_p\Big) = \gamma^2 Q.$$

Equivalently,

$$\Sigma = \gamma^2 \int_0^\infty e^{-s(\gamma A - \frac{1}{2} I_p)}\, Q\, e^{-s(\gamma A^\top - \frac{1}{2} I_p)}\, ds.$$

Finally, for the portfolio estimator

$$u_n := \frac{y_n}{\|y_n\|_1}, \qquad u^\star := \frac{y^\star}{\|y^\star\|_1},$$

one has

$$\sqrt{n}\,(u_n - u^\star) \Rightarrow \mathcal{N}(0, \Sigma_u),$$

where, writing Σ_{yy} *for the lower-right* $d \times d$ *block of* Σ*,*

$$\Sigma_u = B_u\, \Sigma_{yy}\, B_u^\top, \qquad B_u = \frac{1}{\|y^\star\|_1}\Big(I_d - u^\star \mathbf{1}^\top\Big).$$

In terms of the model primitives, S and Q are explicit. If we write, at the random point

$$(\xi^\star, -\langle y^\star, X_1\rangle),$$

$$L_a^\star := \partial_a L(\xi^\star, -\langle y^\star, X_1\rangle), \qquad L_{ab}^\star := \partial_{ab} L(\xi^\star, -\langle y^\star, X_1\rangle),$$

then

$$\nabla_z H(z^\star, X_1) = \begin{pmatrix} L_1^\star \\ -\, L_2^\star X_1 - \big(b_i / y_i^\star\big)_{1 \le i \le d} \end{pmatrix},$$

hence

$$\zeta_1 = \begin{pmatrix} L_1^\star \\ -\,\kappa_\star\, \mathrm{Diag}(y^\star) L_2^\star X_1 - \kappa_\star b \end{pmatrix}, \qquad Q = \mathrm{Cov}(\zeta_1),$$

and

$$S = \mathbb{E}\begin{pmatrix} L_{11}^\star & -L_{12}^\star X_1^\top \\ -L_{12}^\star X_1 & L_{22}^\star X_1 X_1^\top \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \mathrm{Diag}(b_i/(y_i^\star)^2) \end{pmatrix}.$$

Thus the asymptotic covariance is fully determined by (L, X, b), the optimizer $z^\star$, and the mirror/taming factor through $G_\star$.

Proof

Let $\mathcal{F}_n := \sigma(X_1, \dots, X_n)$. All norms are Euclidean/operator norms.

Lemma 1 (Geometry near the interior optimum)**.** *There exists a deterministic neighborhood U of $z^\star$, constants $0 < c_1 < c_2 < m$, and $\varepsilon > 0$ such that for every $z = (\xi, y) \in U$,*

$$\min_{1 \le i \le d} y_i \ge \varepsilon, \qquad \|y\|_1 \le c_2 < m.$$

On U, the matrix-valued map G is bounded and Lipschitz:

$$\sup_{z \in U} \|G(z)\| < \infty, \qquad \|G(z) - G(z')\| \le L_G \|z - z'\| \quad (z, z' \in U)$$

for some deterministic $L_G < \infty$.

Proof. Because $y^\star \in (0, \infty)^d$ and $m > \|y^\star\|_1$, we may choose $\varepsilon > 0$ so small that

$$\min_i y_i^\star > 3\varepsilon, \qquad \|y^\star\|_1 + 3\varepsilon < m.$$

Then take U so small that $z = (\xi, y) \in U$ implies

$$|y_i - y_i^\star| < \varepsilon \ \forall i, \qquad \|y - y^\star\|_1 < \varepsilon.$$

Hence $\min_i y_i \ge 2\varepsilon$ and $\|y\|_1 \le \|y^\star\|_1 + \varepsilon =: c_2 < m$.

Since $\kappa(y) = (\min_i y_i) \wedge 1$ is 1-Lipschitz on $\mathbb{R}^d_+$, the map $y \mapsto \kappa(y)\,\mathrm{Diag}(y)$ is Lipschitz on the compact set $\{y : (\xi, y) \in U\}$. Therefore G is bounded and Lipschitz on U. $\square$

Lemma 2 (Projection is eventually inactive)**.** *Almost surely, the rescaling part of the proximal map $P^m_{y_n}$ is used only finitely many times. Consequently, almost surely, for all sufficiently large n,*

$$y_{n+1} = y_n \odot \exp\Big(-\gamma_{n+1} \kappa(y_n) \nabla_y H(z_n, X_{n+1})\Big).$$

Proof. By almost sure convergence $z_n \to z^\star$, there exists an almost surely finite random index N_1 such that $z_n \in U$ for all $n \ge N_1$.

Fix $n \ge N_1$, and define

$$v_n := \gamma_{n+1} \kappa(y_n) \nabla_y H(z_n, X_{n+1}).$$

Then, by (A4), $\|v_n\|_\infty \le \gamma_{n+1} M_0(X_{n+1})$. Also,

$$\|y_n \odot e^{-v_n}\|_1 \le \|y_n\|_1 e^{\|v_n\|_\infty} \le c_2 e^{\gamma_{n+1} M_0(X_{n+1})}.$$

Choose $c_0 > 0$ so small that $c_2 e^{c_0} < m$. Since $M_0 \in L^3$,

$$\sum_{n \ge 1} \mathbb{P}(\gamma_{n+1} M_0(X_{n+1}) > c_0) \le \sum_{n \ge 1} \frac{\gamma^3 \mathbb{E}[M_0^3]}{c_0^3 (n+1)^3} < \infty.$$

By Borel–Cantelli, almost surely there exists a finite random N_2 such that

$$\gamma_{n+1} M_0(X_{n+1}) \le c_0 \qquad \forall n \ge N_2.$$

Hence, for all $n \ge N := \max(N_1, N_2)$,

$$\|y_n \odot e^{-v_n}\|_1 < m,$$

so the ℓ_1-rescaling part of $P^m_{y_n}$ is inactive. This proves the claim. $\square$

From now on, all estimates are for n large enough that the projection is inactive; finitely many initial steps do not affect the weak limit.

Lemma 3 (One-step mirror expansion)**.** *There exists a deterministic constant $C_q < \infty$ such that, for all sufficiently large n,*

$$z_{n+1} = z_n - \gamma_{n+1}\Psi(z_n, X_{n+1}) + q_{n+1},$$

where $q_{n+1} \in \mathbb{R}^p$ has first coordinate 0 and satisfies

$$\|q_{n+1}\| \le C_q\, \gamma_{n+1}^2\, M_0(X_{n+1})^2.$$

In particular,

$$\mathbb{E}\|q_{n+1}\| = O(n^{-2}), \qquad \mathbb{E}\|q_{n+1}\|^2 = O(n^{-4}).$$

Proof. The ξ-update is exactly

$$\xi_{n+1} = \xi_n - \gamma_{n+1}\partial_\xi H(z_n, X_{n+1}).$$

For the y-update, write componentwise

$$(y_{n+1})_i = (y_n)_i e^{-(v_n)_i}, \qquad (v_n)_i = \gamma_{n+1}\kappa(y_n)\partial_{y_i} H(z_n, X_{n+1}).$$

By Taylor's theorem,

$$e^{-a} = 1 - a + \frac{1}{2}e^{-\theta a}a^2 \qquad \text{for some } \theta \in (0,1).$$

Hence

$$(y_{n+1})_i = (y_n)_i - (y_n)_i(v_n)_i + r_{n+1,i},$$

with

$$|r_{n+1,i}| \le \frac{1}{2}(y_n)_i e^{|(v_n)_i|}\,(v_n)_i^2.$$

For all sufficiently large n, the proof of the previous lemma gives $\|v_n\|_\infty \le c_0$, while $(y_n)_i \le c_2$. Therefore

$$|r_{n+1,i}| \le \frac{1}{2}c_2 e^{c_0}\gamma_{n+1}^2 M_0(X_{n+1})^2.$$

Collecting the d coordinates yields the desired vector remainder bound. □

Lemma 4 (Linearization of the deterministic drift)**.** *Define*

$$F(z) := \mathbb{E}[\Psi(z, X_1)] = G(z)\nabla h(z).$$

Then

$$F(z^\star) = 0, \qquad F(z) = A(z - z^\star) + R_F(z),$$

where

$$A = G_\star S, \qquad \|R_F(z)\| \le C_F\|z - z^\star\|^2 \quad (z \in U)$$

for some deterministic $C_F < \infty$.

Moreover, all eigenvalues of A are real and strictly positive.

Proof. Since $z^\star$ minimizes h, one has $\nabla h(z^\star) = 0$, hence $F(z^\star) = G_\star \nabla h(z^\star) = 0$.

Because $h \in C^2(U)$,

$$\nabla h(z) = S(z - z^\star) + r_h(z), \qquad \|r_h(z)\| \le C\|z - z^\star\|^2.$$

Also, by Lemma 1,

$$\|G(z) - G_\star\| \le L_G\|z - z^\star\|.$$

Therefore

$$F(z) = G(z)\nabla h(z) = G_\star S(z - z^\star) + (G(z) - G_\star)S(z - z^\star) + G(z)r_h(z).$$

The last two terms are $O(\|z - z^\star\|^2)$, so the claimed quadratic remainder follows.

For the spectral claim, note that S is symmetric positive definite and $G_\star$ is diagonal positive definite. Hence

$$A = G_\star S = G_\star^{1/2}\big(G_\star^{1/2} S G_\star^{1/2}\big)G_\star^{-1/2},$$

so A is similar to the symmetric positive-definite matrix $G_\star^{1/2} S G_\star^{1/2}$. Therefore all eigenvalues of A are real and strictly positive. $\square$

Lemma 5 (Decomposition into linear SA plus negligible remainders)**.** *Let*

$$\zeta_{n+1} := \Psi(z^\star, X_{n+1}) = G_\star \nabla_z H(z^\star, X_{n+1}), \qquad Q := \mathbb{E}[\zeta_1\zeta_1^\top].$$

Then $(\zeta_n)_{n\ge1}$ are i.i.d., centered, and have finite third moment.

For all sufficiently large n,

$$e_{n+1} = \Big(I - \frac{\gamma}{n+1}A\Big)e_n - \frac{\gamma}{n+1}\zeta_{n+1} + r_{n+1} + m_{n+1},$$

where:

(i) m_{n+1} is an $\mathcal{F}_{n+1}$-martingale difference:

$$\mathbb{E}[m_{n+1} \mid \mathcal{F}_n] = 0,$$

and

$$\mathbb{E}\big[\|m_{n+1}\|^2 \mid \mathcal{F}_n\big] \le \frac{C_m}{(n+1)^2}\|e_n\|^2;$$

(ii) the deterministic remainder satisfies

$$\mathbb{E}\|r_{n+1}\| \le \frac{C_r}{(n+1)^2}, \qquad \mathbb{E}\|r_{n+1}\|^2 \le \frac{C_r}{(n+1)^4}.$$

Proof. Set

$$\Delta_{n+1} := \Psi(z_n, X_{n+1}) - F(z_n).$$

Then $\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] = 0$, and the recursion of the previous lemma can be rewritten as

$$e_{n+1} = e_n - \gamma_{n+1}F(z_n) - \gamma_{n+1}\Delta_{n+1} + q_{n+1}.$$

Add and subtract Ae_n and ζ_{n+1}:

$$e_{n+1} = \Big(I - \frac{\gamma}{n+1}A\Big)e_n - \frac{\gamma}{n+1}\zeta_{n+1} + r_{n+1} + m_{n+1},$$

where

$$r_{n+1} := -\frac{\gamma}{n+1}\big(F(z_n) - Ae_n\big) + q_{n+1},$$

$$m_{n+1} := -\frac{\gamma}{n+1}\big(\Delta_{n+1} - \zeta_{n+1}\big).$$

The vector ζ_{n+1} is centered because

$$\mathbb{E}[\zeta_{n+1}] = \mathbb{E}[\Psi(z^\star, X_{n+1})] = F(z^\star) = 0.$$

Its finite third moment follows from $\|\zeta_{n+1}\| \le \|G_\star\| M_0(X_{n+1})$ and $M_0 \in L^3$.

For r_{n+1}, Lemma 4 gives $\|F(z_n) - Ae_n\| \le C_F \|e_n\|^2$, so

$$\mathbb{E}\Big\|\frac{\gamma}{n+1}\big(F(z_n) - Ae_n\big)\Big\| \le \frac{\gamma C_F}{n+1}\mathbb{E}\|e_n\|^2 \le \frac{C}{(n+1)^2}$$

by (A5). Combining this with the bound on q_{n+1} yields the stated estimates for r_{n+1}.

For m_{n+1}, define

$$\widetilde{M}_1 := \|G\|_{\infty,U} M_1 + L_G M_0,$$

where $\|G\|_{\infty,U} := \sup_{z\in U} \|G(z)\|$. Then $\widetilde{M}_1 \in L^3$, and for all $z, z' \in U$,

$$\|\Psi(z, X_1) - \Psi(z', X_1)\| \le \widetilde{M}_1 \|z - z'\|.$$

Hence

$$\|\Delta_{n+1} - \zeta_{n+1}\| = \Big\|[\Psi(z_n, X_{n+1}) - \Psi(z^\star, X_{n+1})] - [F(z_n) - F(z^\star)]\Big\| \le 2\widetilde{M}_1(X_{n+1})\|e_n\|.$$

Therefore

$$\mathbb{E}\big[\|m_{n+1}\|^2 \mid \mathcal{F}_n\big] \le \frac{4\gamma^2}{(n+1)^2}\mathbb{E}[\widetilde{M}_1(X_1)^2]\|e_n\|^2.$$

This proves the claim. □

Lemma 6 (Asymptotics of the linear matrix product)**.** *Let*

$$\Pi_{k,n} := \prod_{j=k}^{n}\Big(I - \frac{\gamma}{j}A\Big), \qquad \Pi_{n+1,n} := I.$$

Then there exists $C_\Pi < \infty$ such that for all $1 \le k \le n$,

$$\|\Pi_{k+1,n}\| \le C_\Pi \Big(\frac{k}{n}\Big)^\beta.$$

Moreover, if $t^M := e^{(\log t)M}$ for $0 < t \le 1$, then

$$\Pi_{k+1,n} = \Big(\frac{k}{n}\Big)^{\gamma A}\big(I + E_{k,n}\big), \qquad \sup_{n\ge k}\|E_{k,n}\| \le \frac{C_\Pi}{k}$$

for all sufficiently large k.

Proof. All factors $I - \gamma A/j$ are polynomials in the fixed matrix A, hence commute. Choose j_0 large so that $\|\gamma A/j\| \le 1/2$ for $j \ge j_0$. For such j, the principal matrix logarithm is defined and

$$\log\Big(I - \frac{\gamma}{j}A\Big) = -\frac{\gamma}{j}A + R_j, \qquad \|R_j\| \le \frac{C}{j^2}.$$

Hence, for $n \ge k \ge j_0$,

$$\Pi_{k+1,n} = \exp\Big(-\gamma A \sum_{j=k+1}^{n} \frac{1}{j} + \sum_{j=k+1}^{n} R_j\Big).$$

Since

$$\sum_{j=k+1}^{n} \frac{1}{j} = \log\frac{n}{k} + \eta_{k,n}, \qquad |\eta_{k,n}| \le \frac{C}{k},$$

and

$$\Big\|\sum_{j=k+1}^{n} R_j\Big\| \le \frac{C}{k},$$

we obtain

$$\Pi_{k+1,n} = \Big(\frac{k}{n}\Big)^{\gamma A} \exp\Big(-\gamma A\, \eta_{k,n} + \sum_{j=k+1}^{n} R_j\Big).$$

The exponential factor is $I + O(1/k)$, uniformly in $n \ge k$, proving the second claim.

For the first claim, diagonalize $A = PDP^{-1}$ with $D = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_i > 0$. Then

$$\Big\|\Big(\frac{k}{n}\Big)^{\gamma A}\Big\| \le \|P\|\|P^{-1}\| \max_i \Big(\frac{k}{n}\Big)^{\gamma\lambda_i} \le C\Big(\frac{k}{n}\Big)^{\beta}.$$

The finitely many cases $k < j_0$ are absorbed into the constant. □

Lemma 7 (Reduction to the linear recursion)**.** *Let $(\widetilde{e}_n)$ be defined by*

$$\widetilde{e}_{n+1} = \Big(I - \frac{\gamma}{n+1}A\Big)\widetilde{e}_n - \frac{\gamma}{n+1}\zeta_{n+1}, \qquad \widetilde{e}_0 = e_0.$$

Then

$$\sqrt{n}\,(e_n - \widetilde{e}_n) \xrightarrow{\mathbb{P}} 0.$$

Proof. Set $d_n := e_n - \widetilde{e}_n$. By the previous lemma,

$$d_{n+1} = \Big(I - \frac{\gamma}{n+1}A\Big)d_n + r_{n+1} + m_{n+1}, \qquad d_0 = 0.$$

Therefore

$$d_n = \sum_{k=1}^{n} \Pi_{k+1,n}(r_k + m_k).$$

For the deterministic remainder,

$$\sqrt{n}\,\mathbb{E}\Big\|\sum_{k=1}^{n} \Pi_{k+1,n} r_k\Big\| \le \sqrt{n}\sum_{k=1}^{n} \|\Pi_{k+1,n}\|\mathbb{E}\|r_k\| \le C n^{1/2-\beta} \sum_{k=1}^{n} k^{\beta-2}.$$

If $\beta < 1$, the series $\sum k^{\beta-2}$ converges, so the right-hand side is $O(n^{1/2-\beta}) \to 0$. If $\beta \geq 1$, then $\sum_{k \leq n} k^{\beta-2} = O(n^{\beta-1})$, so the bound is $O(n^{-1/2}) \to 0$. Hence

$$\sqrt{n} \sum_{k=1}^{n} \Pi_{k+1,n} r_k \to 0 \quad \text{in } L^1.$$

For the martingale remainder, orthogonality of the martingale differences m_k yields

$$\mathbb{E}\Big\| \sqrt{n} \sum_{k=1}^{n} \Pi_{k+1,n} m_k \Big\|^2 = n \sum_{k=1}^{n} \mathbb{E}\|\Pi_{k+1,n} m_k\|^2.$$

Using the previous lemma and the conditional variance bound on m_k,

$$\mathbb{E}\Big\| \sqrt{n} \sum_{k=1}^{n} \Pi_{k+1,n} m_k \Big\|^2 \leq Cn \sum_{k=1}^{n} \Big(\frac{k}{n}\Big)^{2\beta} \frac{1}{k^2} \mathbb{E}\|e_{k-1}\|^2.$$

By (A5), $\mathbb{E}\|e_{k-1}\|^2 \leq C/k$, so

$$\mathbb{E}\Big\| \sqrt{n} \sum_{k=1}^{n} \Pi_{k+1,n} m_k \Big\|^2 \leq Cn^{1-2\beta} \sum_{k=1}^{n} k^{2\beta-3}.$$

If $\beta < 1$, the series converges; if $\beta \geq 1$, it grows like $O(n^{2\beta-2})$. In either case, because $\beta > 1/2$, the whole bound tends to 0.

Thus

$$\sqrt{n} \sum_{k=1}^{n} \Pi_{k+1,n} m_k \to 0 \quad \text{in } L^2,$$

hence in probability. Combining the two parts proves the claim. □

Lemma 8 (CLT for the linearized recursion)**.** *One has*

$$\sqrt{n}\, \widetilde{e}_n \Rightarrow \mathcal{N}(0, \Sigma),$$

where

$$\Sigma = \gamma^2 \int_0^\infty e^{-s(\gamma A - \frac{1}{2} I_p)} Q\, e^{-s(\gamma A^\top - \frac{1}{2} I_p)}\, ds.$$

Proof. The explicit solution of the linear recursion is

$$\widetilde{e}_n = \Pi_{1,n} \widetilde{e}_0 - \sum_{k=1}^{n} \frac{\gamma}{k} \Pi_{k+1,n} \zeta_k.$$

The initial term is negligible because

$$\|\sqrt{n}\, \Pi_{1,n} \widetilde{e}_0\| \leq Cn^{1/2-\beta} \|\widetilde{e}_0\| \to 0.$$

Define

$$D_{k,n} := \frac{\gamma\sqrt{n}}{k} \Pi_{k+1,n}, \qquad Y_{k,n}(u) := u^\top D_{k,n} \zeta_k \quad (u \in \mathbb{R}^p).$$

Then

$$u^\top \sqrt{n}\, \widetilde{e}_n = -\sum_{k=1}^{n} Y_{k,n}(u) + o(1).$$

Since the ζ_k are i.i.d. and centered, the scalar variables $Y_{k,n}(u)$ are independent and centered.

Step 1: bounds on the coefficients. From the product lemma,

$$\|D_{k,n}\| \le Cn^{-1/2}\Big(\frac{k}{n}\Big)^{\beta-1}.$$

Hence

$$\max_{1\le k\le n} \mathbb{E}[Y_{k,n}(u)^2] \le \|u\|^2\mathbb{E}\|\zeta_1\|^2 \max_{k\le n}\|D_{k,n}\|^2 \to 0$$

because $\beta > 1/2$.

Also,

$$\sum_{k=1}^n \mathbb{E}|Y_{k,n}(u)|^3 \le \|u\|^3\mathbb{E}\|\zeta_1\|^3 \sum_{k=1}^n \|D_{k,n}\|^3.$$

Now

$$\sum_{k=1}^n \|D_{k,n}\|^3 \le Cn^{-3/2}\sum_{k=1}^n\Big(\frac{k}{n}\Big)^{3\beta-3}.$$

If $3\beta-3<-1$, the sum in k is bounded and the right-hand side is $O(n^{3/2-3\beta}) \to 0$. If $3\beta-3 \ge -1$, the sum is $O(n^{3\beta-2})$, so the right-hand side is $O(n^{-1/2}) \to 0$. Therefore

$$\sum_{k=1}^n \mathbb{E}|Y_{k,n}(u)|^3 \to 0.$$

Step 2: covariance limit. Set

$$\Sigma_n := \sum_{k=1}^n D_{k,n}QD_{k,n}^\top.$$

Using the refined product asymptotic,

$$D_{k,n} = \frac{\gamma}{\sqrt{n}}\Big(\frac{k}{n}\Big)^{\gamma A - I_p}(I+E_{k,n}), \qquad \|E_{k,n}\| \le \frac{C}{k}.$$

Hence

$$\Sigma_n = \frac{\gamma^2}{n}\sum_{k=1}^n\Big(\frac{k}{n}\Big)^{\gamma A-I_p}Q\Big(\frac{k}{n}\Big)^{\gamma A^\top-I_p} + R_n,$$

where

$$\|R_n\| \le C\frac{1}{n}\sum_{k=1}^n\frac{1}{k}\Big(\frac{k}{n}\Big)^{2\beta-2} = Cn^{1-2\beta}\sum_{k=1}^n k^{2\beta-3} \to 0.$$

Define

$$f(t) := \gamma^2 t^{\gamma A-I_p}Qt^{\gamma A^\top-I_p}, \qquad 0<t\le 1.$$

Because A is diagonalizable with positive spectrum,

$$\|f(t)\| \le Ct^{2\beta-2}.$$

Since $2\beta-2>-1$, f is integrable near 0. Therefore the Riemann sums converge:

$$\frac{1}{n}\sum_{k=1}^n f(k/n) \to \int_0^1 f(t)\,dt.$$

So $\Sigma_n \to \Sigma$, where

$$\Sigma = \gamma^2 \int_0^1 t^{\gamma A - I_p} Q t^{\gamma A^\top - I_p}\, dt.$$

Now substitute $t = e^{-s}$. Reversing the limits and splitting the Jacobian factor $dt = t\,ds$ symmetrically,

$$\Sigma = \gamma^2 \int_0^\infty e^{-s(\gamma A - \frac{1}{2} I_p)} Q e^{-s(\gamma A^\top - \frac{1}{2} I_p)}\, ds.$$

Step 3: characteristic functions. Fix $u \in \mathbb{R}^p$. Since $(Y_{k,n}(u))$ are independent,

$$\mathbb{E} \exp\Big(i \sum_{k=1}^n Y_{k,n}(u)\Big) = \prod_{k=1}^n \mathbb{E}[e^{iY_{k,n}(u)}].$$

For any centered scalar random variable Y with finite third moment,

$$e^{iY} = 1 + iY - \frac{1}{2} Y^2 + \rho(Y), \qquad |\rho(Y)| \le \frac{|Y|^3}{6},$$

hence

$$\mathbb{E}[e^{iY}] = 1 - \frac{1}{2}\mathbb{E}[Y^2] + \theta, \qquad |\theta| \le \frac{1}{6}\mathbb{E}|Y|^3.$$

Apply this to $Y = Y_{k,n}(u)$. Because $\max_k \mathbb{E}[Y_{k,n}(u)^2] \to 0$, for large n all factors are uniformly close to 1, and

$$\log \mathbb{E}[e^{iY_{k,n}(u)}] = -\frac{1}{2}\mathbb{E}[Y_{k,n}(u)^2] + O\Big(\mathbb{E}|Y_{k,n}(u)|^3 + \mathbb{E}[Y_{k,n}(u)^2]^2\Big).$$

Summing over k and using

$$\sum_{k=1}^n \mathbb{E}|Y_{k,n}(u)|^3 \to 0, \qquad \sum_{k=1}^n \mathbb{E}[Y_{k,n}(u)^2]^2 \le \Big(\max_k \mathbb{E}[Y_{k,n}(u)^2]\Big) \sum_{k=1}^n \mathbb{E}[Y_{k,n}(u)^2] \to 0,$$

we obtain

$$\log \mathbb{E} \exp\Big(i \sum_{k=1}^n Y_{k,n}(u)\Big) \to -\frac{1}{2} u^\top \Sigma u.$$

Therefore

$$u^\top \sqrt{n}\, \widetilde{e}_n \Rightarrow \mathcal{N}(0, u^\top \Sigma u) \qquad \forall u \in \mathbb{R}^p.$$

By the Cramér–Wold device,

$$\sqrt{n}\, \widetilde{e}_n \Rightarrow \mathcal{N}(0, \Sigma).$$

□

We can now finish the theorem.

Proof of the theorem. By the previous two lemmas,

$$\sqrt{n}\, e_n = \sqrt{n}\, \widetilde{e}_n + \sqrt{n}\,(e_n - \widetilde{e}_n),$$

with

$$\sqrt{n}\, \widetilde{e}_n \Rightarrow \mathcal{N}(0, \Sigma), \qquad \sqrt{n}\,(e_n - \widetilde{e}_n) \xrightarrow{\mathbb{P}} 0.$$

Slutsky's theorem gives

$$\sqrt{n}\,(z_n - z^\star) = \sqrt{n}\, e_n \Rightarrow \mathcal{N}(0, \Sigma).$$

It remains to identify Σ as the unique solution of the Lyapunov equation. Let

$$B := \gamma A - \frac{1}{2} I_p.$$

Since $\min \Re\lambda(B) = \beta - \frac{1}{2} > 0$, the integral

$$\Sigma = \gamma^2 \int_0^\infty e^{-sB} Q e^{-sB^\top} \, ds$$

is finite. Moreover,

$$\frac{d}{ds}\big(e^{-sB} Q e^{-sB^\top}\big) = -\, B e^{-sB} Q e^{-sB^\top} - e^{-sB} Q e^{-sB^\top} B^\top.$$

Integrating from 0 to ∞ gives

$$B\Sigma + \Sigma B^\top = \gamma^2 Q.$$

For uniqueness, suppose X solves $BX + XB^\top = 0$. Then

$$\frac{d}{ds}\big(e^{-sB} X e^{-sB^\top}\big) = 0,$$

so $e^{-sB} X e^{-sB^\top}$ is constant in s. Since B is stable, the left-hand side converges to 0 as $s \to \infty$, hence the constant is 0, so $X = 0$.

Finally, consider the map $\pi(y) = y/\|y\|_1$. Since $y^\star \in (0,\infty)^d$, π is C^1 near $y^\star$, with derivative

$$D\pi(y^\star)[v] = \frac{1}{\|y^\star\|_1} \Big(I_d - u^\star \mathbf{1}^\top \Big) v = B_u v.$$

Therefore

$$u_n - u^\star = B_u (y_n - y^\star) + o(\|y_n - y^\star\|).$$

Because $\sqrt{n}(y_n - y^\star)$ converges in distribution, the remainder is $o_{\mathbb{P}}(n^{-1/2})$. Hence

$$\sqrt{n}\,(u_n - u^\star) = B_u \sqrt{n}\,(y_n - y^\star) + o_{\mathbb{P}}(1) \Rightarrow \mathcal{N}(0, B_u \Sigma_{yy} B_u^\top).$$

This proves the theorem. □

Gap to the full statement

What remains unproved is precise and substantial:

1. I did *not* prove a CLT for the paper's weighted average

$$\bar{z}_n = \frac{\sum_{k=1}^n \gamma_k z_{k-1}}{\sum_{k=1}^n \gamma_k}.$$

   That average is not the classical Polyak–Ruppert average, and its asymptotic scaling is genuinely different.

2. I did *not* treat the boundary case $m = \|y^\star\|_1$, or any situation in which the ℓ_1-projection remains asymptotically active. In such cases one expects a reflected linear diffusion rather than the unconstrained Gaussian law above.

3. I imposed a stronger first-order rate assumption,

$$\sup_n n\,\mathbb{E}\|z_n - z^\star\|^2 < \infty,$$

which the prompt does not explicitly provide. It is standard in strongly stable SA, but I did not derive it here from the SMD recursion itself.

4. I restricted to $\gamma_n = \gamma/n$. I did not analyze $\gamma_n \asymp n^{-\alpha}$ with $\alpha \in (1/2, 1)$, nor the asymptotics of the prompt's weighted average under such stepsizes.

So the result is a rigorous *interior raw-iterate CLT*, plus the induced portfolio-weight CLT, not a full solution for all versions of the prompt.

Self red-team

Counterexample attempts.

1. *Boundary-active case.* If $m = \|y^\star\|_1$, then even in linear constrained SA one generally gets a reflected diffusion rather than the unconstrained Gaussian covariance above. This is exactly the failure mode ruled out by (A2).

2. *Spectral threshold.* In the scalar linear recursion

$$x_{n+1} = \Big(1 - \frac{a}{n+1}\Big)x_n + \frac{\xi_{n+1}}{n+1}, \qquad \mathbb{E}[\xi_n] = 0,\ \operatorname{Var}(\xi_n) = 1,$$

one computes

$$\operatorname{Var}(\sqrt{n}\,x_n) \asymp n^{1-2a}\sum_{k=1}^{n} k^{2a-2},$$

which stays bounded iff $a > 1/2$. Thus the condition $\min \Re\lambda(\gamma A) > 1/2$ is not cosmetic.

3. *Degenerate one-asset case $d = 1$.* Then $u_n \equiv 1$ identically. My portfolio-weight covariance formula gives

$$B_u = \frac{1}{\|y^\star\|_1}(1 - u^\star \cdot 1) = 0,$$

hence $\Sigma_u = 0$, exactly as it should.

Three plausible referee objections.

1. *"κ is not differentiable; how can the drift be linearized?"* I never differentiate κ. The proof only uses that $G(z) = \operatorname{diag}(1, \kappa(y)\operatorname{Diag}(y))$ is Lipschitz near $z^\star$. Because $\nabla h(z^\star) = 0$, the first-order term is $G_\star \nabla^2 h(z^\star)$, and the variation of G contributes only a quadratic remainder.

2. *"The projection may still fire infinitely often because the gradient is random."* This is addressed in the proof of eventual inactivity via Borel–Cantelli. The third-moment envelope and the $1/n$-stepsize imply

$$\sum_n \mathbb{P}(\gamma_n M_0(X_n) > c_0) < \infty,$$

so only finitely many large jumps occur almost surely.

3. *"You froze the noise at $z^\star$; state-dependent noise could change the covariance."* I did not ignore it. The term m_{n+1} is precisely the state-dependent martingale remainder. Its cumulative $\sqrt{n}$-scale variance is shown to vanish:

$$\mathbb{E}\left\|\sqrt{n}\sum_{k=1}^{n}\Pi_{k+1,n}m_k\right\|^2 \to 0.$$

So it is asymptotically negligible.

Boundary and degenerate checks. I checked:

- the scalar $d = 1$ portfolio limit, which collapses correctly to a point mass;
- the threshold $\beta > 1/2$, which is sharp already in the scalar linear model;
- the cases $\kappa_\star = 1$ and $\kappa_\star < 1$, both covered uniformly since only $G_\star$ enters the limit law.

Ideas tried

1. **Weighted-average CLT via Polyak–Ruppert.** Idea: identify $\bar{z}_n$ with a trajectory average and invoke the classical averaged-SA CLT. Obstruction: the paper's average is $\sum \gamma_k z_{k-1}/\sum \gamma_k$, not the usual Cesàro average $(1/n)\sum z_k$. Its asymptotic scaling is different, and in fact for $\gamma_k = \gamma/k$ the early iterates are weighted too heavily for a naive Polyak–Ruppert theorem to apply. Natural next step: derive a dedicated Abel-transform analysis for the γ_k-weighted average.

2. **Direct black-box application of Fabian/Hernandez–Spall.** Idea: reduce the SMD recursion to a standard Robbins–Monro form and quote a raw-iterate SA CLT. Obstruction: this would hide the most interesting part of the problem, namely how the KL mirror step and the taming factor κ modify the Jacobian and the noise covariance. It also leaves the projection issue opaque. Natural next step: now that the explicit reduction is proved, one could repackage the result under an audited abstract SA theorem.

3. **Log-coordinate transform $s_i = \log y_i$.** Idea: when the projection is inactive, the entropic mirror step is additive in s-coordinates:

$$s_{n+1} = s_n - \gamma_{n+1}\kappa(e^{s_n})\nabla_y H(z_n, X_{n+1}).$$

Obstruction: although this simplifies the mirror step, the portfolio map $y = e^s$ and the eventual normalization $u = y/\|y\|_1$ become less transparent, and the interaction with the radius truncation is not simpler. Natural next step: use the log-chart to study moderate deviations or higher-order asymptotics.

4. **Full boundary case $m = \|y^\star\|_1$.** Idea: keep the projection active and identify the normalized limit as a reflected diffusion. Obstruction: this requires the full machinery of constrained SA weak convergence, including an identification of the effective reflection field generated by the KL projection on the ℓ_1-ball. Natural next step: combine the linearization carried out here with constrained-SA reflected-diffusion techniques.

Invoked results

None as proof steps. I intentionally avoided importing a black-box stochastic-approximation CLT. The Gaussian limit above is proved directly from:

- a first-order reduction of the mirror recursion to a linear SA recursion;
- explicit asymptotics for the deterministic matrix product $\prod_{j=k}^{n}(I - \gamma A/j)$;
- a characteristic-function argument for the resulting independent triangular array.

The literature listed earlier was used only for orientation and for identifying the expected covariance/Lyapunov form and the boundary-active obstruction.

Self-confidence

3

I checked every nontrivial step and deliberately attacked the main weak points (projection activity, nonsmooth κ, and state-dependent noise), but the result is still a strong partial theorem with extra assumptions rather than the full general statement.

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